

Java OOP Implementation Assignment

Project Title

Smart Library & Book Management System

Problem Statement

A university library wants to replace its manual book management process with a software application. Develop a menu-driven console-based Java application that automates library operations using Object-Oriented Programming principles. The application should be modular, reusable, and easy to extend.

Functional Requirements

Book Management

- Add a new book.
- Display all books.
- Update book details.
- Delete a book using Book ID.

Search Operations

- Search by Book ID.
- Search by Title.
- Search by Author.
- Search by Category.

Library Operations

- Issue a book.
- Return a book.
- Display available books.
- Display issued books.
- Display total, available and issued book counts.

Book Information

Book ID, Book Title, Author Name, Publisher Name, ISBN Number, Category, Price, Number of Pages, Publication Year and Availability Status.

Printed Book: Shelf Number, Edition.

E-Book: File Format, File Size, Download Link.

Mandatory OOP Concepts

- Classes and Objects
- Encapsulation using private data members and getters/setters
- Default and Parameterized Constructors
- Constructor Overloading
- Inheritance
- Method Overloading
- Method Overriding
- Abstract Class
- Interface
- Static Variables and Methods
- Arrays of Objects (Collections Framework is not allowed)
- Composition

Validation Rules

- Book ID must be unique.
- ISBN Number must be unique.
- Price cannot be negative.
- Number of Pages must be greater than zero.
- A book cannot be issued if it is already issued.
- Returning a book should change its status to available.
- Book ID cannot be modified during update.
- Display an appropriate message if a searched book does not exist.

Menu

1. Add Book
2. Display All Books
3. Search Book
4. Update Book
5. Delete Book
6. Issue Book
7. Return Book
8. Display Available Books
9. Display Issued Books
10. Display Library Statistics
11. Exit

Implementation Constraints

- Use Java only.
- Use Arrays of Objects only.
- Collections Framework is not allowed.
- Exception Handling is not allowed.
- File Handling is not allowed.
- Database Connectivity is not allowed.
- GUI is not allowed.
- Follow proper naming conventions and modular programming practices.

Expected Deliverables

- Complete Java source code.
- Menu-driven application.
- Proper implementation of all mandatory OOP concepts.
- Meaningful output for every operation.

Evaluation Criteria

Criteria	Marks
Program Design	15
OOP Concepts	30
Menu Implementation	15
Search, Update & Delete	15
Issue & Return Functionality	10
Code Quality	10

